Course name: Introduction to Machine Learning, Spring 2025

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Machine Learning Assignment

Introduction

The Individual Take-home Assignment 2 is worth 100 points (20% of your final grade). While a passing grade on this assignment is not required provided the overall course grade is at least 5.5, no resit is available for the assignment as a whole. Students who fail may resubmit for a maximum grade of 6.0, provided the re-submission is received before the final exam.

Late Submission Policy:

Late submissions incur a 5% penalty per day. Submissions received more than 7 days past the due date will receive zero credit. Start early to avoid penalties.

Submission Requirements:

- Submit your code as a Jupyter Notebook (`*.ipynb`), including all results and plots.
- Additional functions may be provided in a separate Python file (`*.py`).
- Include references at the top of your notebook for any external code, methods, or ideas not originally developed for this course (excluding course materials such as worksheets, textbook, or lecture slides).
- Provide any special instructions needed to run your code.

Report Requirements:

Submit a report (maximum 3 pages, excluding references and appendix) in PDF format. The report should detail your methods, experiments, and results, and be based on the Jupyter Notebook (no data files should be included). Ensure your code is well-commented so that its correspondence with the report is clear.

Grading Criteria:

Grades will be based on the quality of your work as demonstrated by both your report and your code.

Data Set Information:

The dataset contains profile data from a popular social media platform for a binary classification task to distinguish between automated accounts (bots) and genuine users. It includes numerical features (e.g., activity metrics, follower counts) and categorical features (e.g., profile photo presence, privacy settings). Missing numerical values are represented as `NaN`, while missing categorical values are labeled as "unknown." The target variable `target` is defined as 1 for bots and 0 for genuine users.

Report

Your report (maximum 3 pages, PDF) should include:

- Your name and student number.
- A description of your experiments and results, covering:
  - Features used or omitted.
  - The learning models and algorithms employed.
  - Hyperparameter tuning.
  - The method or system developed for classification.
  - A discussion of experiments (e.g., confusion matrices, per-class accuracy, etc.).
  - References and/or appendices (visualizations, flowcharts, tables) as needed.

Refer to the Grading section for further guidance on structure.

Submission

Submit your assignment via Canvas. Your submission must include:

- The Report (`.pdf`)
- The Code (`.ipynb`)

Your code should clearly mirror the experiments and tuning described in the report. Incomplete submissions (code without report or vice versa) will not be evaluated.

Academic Integrity: Collaboration or code sharing is strictly prohibited. All submissions will be checked for plagiarism, and violations will be reported to the Board of Examiners.

Allowed Resources:

- Code examples provided during the course.
- Open source Python libraries.
- Publicly available open source code from Github (must be credited with a link to the source).

Assignment Details

Binary Classification Task

Reserve 20% of the data as a hold-out test set (use random state 42 for reproducibility). Train two models of your choice (e.g., Neural Networks, Logistic Regression, SVM, KNN) and compare their performance. For each model, perform hyperparameter tuning and evaluate the best configuration on the test set. Your complete pipeline must include training, validation, and testing phases.

Methodology

Provide a high-level overview of your approach, including:

- The rationale for selecting your two models.
- Identification of a baseline model with justification.
- A comparative analysis of the two models.

Constraints:

- Your solution must be fully automated; re-running your code should regenerate predictions without manual intervention.
- Your code must be generic and adaptable to similar datasets without modifications.
- All software components must be open-source and locally installable. Proprietary software or external web services are not permitted.
- Do not incorporate external datasets that overlap with the provided data (instructor approval is required for external data).

Suggested Structure and Grading Criteria

Section 1: Models (15 points)

- Provide detailed descriptions and justifications for your model choices.
- Compare the models concisely and explain your baseline selection.

Section 2: Data Preparation (10 points)

- Describe the dataset and preprocessing steps.
- Explain any feature engineering or justify its omission.
- Detail your data splitting strategy for training, validation, and testing.

Section 3: Model Training (35 points)

- List the hyperparameters tuned for each model along with candidate values.
- Discuss how hyperparameter choices affect performance.
- Specify the final model configurations, including any regularization and training details.

Section 4: Results and Discussion (40 points)

- **Performance Evaluation (10 points):** Report metrics (accuracy, precision, recall, F1-score, AUC) and include confusion matrices, ROC, and precision-recall curves.
- **Model Comparison and Generalization (10 points):** Compare performance on the test set using quantitative metrics and discuss generalizability.
- **Feature Analysis and Group Comparison (10 points):** Describe any feature selection or engineering, identify key features (e.g., via importance scores or SHAP values), and analyze error disparities across the `gender` variable.
- **Critical Reflection (10 points):** Assess the strengths and weaknesses of your approach, and discuss transparency, reproducibility, and potential improvements.